Internship Report on

“ANOMALY DETECTION IN ELECTROCARDIOGRAM (ECG)”

Bachelor of Technology in

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Abstract

This study focuses on developing an automated system for classifying 12-lead electrocardiogram (ECG) signals to detect arrhythmias. We utilized a deep learning approach, specifically a Residual Network (ResNet) model, to effectively capture intricate patterns within ECG data. The dataset comprises recordings from a large-scale ECG database, which were preprocessed and labeled as either normal or anomalous based on their diagnostic annotations. Since the dataset does not explicitly provide diagnosis names but instead uses corresponding codes, we identified atrial fibrillation cases as anomalies based on their associated diagnostic codes. The ECG signals underwent padding and reshaping to ensure uniform input dimensions for the model. To address class imbalance, class weights were computed and applied during model training. The ResNet model was trained using the processed dataset, with early stopping implemented to prevent overfitting. The system's performance was evaluated using metrics such as accuracy, precision, recall, and confusion matrix. The findings suggest that deep learning models, particularly ResNet architectures, are effective in classifying ECG signals and have the potential to enhance automated arrhythmia detection.

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01. Introduction

Electrocardiogram (ECG) anomaly detection is a critical component in modern healthcare, serving as a non-invasive method to monitor and diagnose cardiac conditions. The ECG records the heart's electrical activity, providing essential insights into heart rate, rhythm, and the presence of abnormalities. Accurate detection of anomalies within ECG signals is vital for early diagnosis and effective treatment of various cardiac diseases.

Importance of ECG Anomaly Detection

ECG anomaly detection plays a pivotal role in identifying irregular heart activities, such as arrhythmias, ischemic changes, and other cardiac disorders. Early detection of these anomalies enables timely medical interventions, potentially preventing severe complications like heart attacks or strokes. Moreover, continuous ECG monitoring can aid in assessing the efficacy of treatments and managing chronic heart conditions.

Overview of ECG Signals and Their Clinical Relevance

An ECG signal represents the heart's electrical impulses as a time series, typically captured through multiple leads placed on the body. Each segment of the ECG waveform corresponds to specific cardiac events, such as atrial and ventricular depolarization and repolarization. Clinicians analyze these waveforms to detect abnormalities in heart function. For instance, deviations in the P-wave, QRS complex, or T-wave can indicate various cardiac issues, making ECG an indispensable tool in clinical diagnostics.

Challenges in Anomaly Detection

Detecting anomalies in ECG signals presents several challenges:

- Variability Among Individuals: ECG patterns can vary significantly between individuals due to factors like age, gender, and underlying health conditions, complicating the establishment of universal detection criteria.
- Complexity of ECG Signals: The intricate nature of ECG waveforms, with subtle variations distinguishing normal from abnormal patterns, requires sophisticated analysis techniques.
- Data Quality Issues: ECG recordings are susceptible to noise and artifacts from patient movement or equipment interference, which can obscure critical features necessary for accurate anomaly detection.
- Imbalanced Data: In clinical datasets, normal heartbeats often outnumber abnormal ones, posing challenges for training models that can effectively detect rare anomalies.

Objectives of the Report

This report aims to:

1. **Develop an Automated ECG Anomaly Detection System**: Implement a machine learning-based approach to accurately identify anomalies in ECG signals, enhancing early diagnosis of cardiac conditions.
2. **Address Signal Variability and Noise**: Incorporate preprocessing techniques to mitigate the effects of noise and individual variability, improving the robustness of the detection system.
3. **Handle Data Imbalance**: Employ strategies such as class weighting to manage imbalanced datasets, ensuring the model maintains sensitivity to both normal and abnormal heartbeats.
4. **Evaluate Model Performance**: Assess the system's accuracy, precision, recall, and other relevant metrics using a comprehensive dataset to validate its effectiveness in real-world scenarios.
5. **Facilitate Real-Time Application**: Design the detection algorithm with computational efficiency in mind, enabling its deployment in real-time monitoring systems for continuous cardiac health assessment.

02. Literature Review

Overview of Existing Methods in ECG Anomaly Detection

Electrocardiogram (ECG) anomaly detection has been a focal point in biomedical engineering, aiming to identify irregularities in heart rhythms that may indicate cardiac conditions. Traditional methods primarily relied on manual interpretation by clinicians, which, while effective, are time-consuming and subject to inter-observer variability. To enhance accuracy and efficiency, various automated techniques have been developed over the years.

Traditional Signal Processing Techniques: Early automated methods employed techniques such as Fourier transforms and wavelet analysis to extract features from ECG signals. These features were then used in conjunction with classifiers like Support Vector Machines (SVMs) and k-Nearest Neighbors (k-NN) to detect anomalies. While these approaches provided a foundation for automated ECG analysis, they often struggled with generalization due to the variability in ECG patterns among different individuals.

Machine Learning Approaches: With advancements in computational power, machine learning algorithms became prevalent in ECG anomaly detection. Techniques such as Artificial Neural Networks (ANNs) and Decision Trees were utilized to model complex patterns in ECG data. However, these models required extensive feature engineering and were limited by their capacity to handle large datasets effectively.

Deep Learning Models: The advent of deep learning revolutionized ECG anomaly detection by enabling models to automatically learn hierarchical features from raw data. Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have been extensively applied, demonstrating superior performance in capturing temporal and spatial dependencies in ECG signals. For instance, Long Short-Term Memory (LSTM) networks have been employed to detect anomalies in ECG streams, leveraging their ability to model long-term dependencies.

IEEE Xplore

Advantages of Deep Learning Models, Particularly ResNet

Deep learning models offer several advantages over traditional methods in ECG anomaly detection:

- *Automatic Feature Extraction*: Deep learning models can automatically learn relevant features from raw ECG data, eliminating the need for manual feature engineering.
- *Handling Data Variability*: These models are adept at managing the inherent variability in ECG signals across different populations, improving generalization.
- *Scalability*: Deep learning models can be trained on large datasets, enhancing their robustness and accuracy.

Residual Networks (ResNets), a type of deep learning architecture, have shown particular promise due to their unique design, which addresses the vanishing gradient problem in deep networks. By incorporating shortcut connections that bypass one or more layers, ResNets facilitate the training of very deep networks, allowing for the modeling of complex patterns in ECG data. This architecture has been successfully applied in various medical image processing tasks, including the classification of lung tumors, breast cancer, and skin conditions.

PubMed

03. Dataset Description

Source and Structure of the Dataset

The dataset utilized in this study is sourced from the WFDBRecords repository, a comprehensive collection of ECG recordings stored in a structured directory format. Each record comprises multiple files, including header files (`.hea`), signal files (`.dat`), and annotation files (`.atr`), organized systematically to facilitate efficient data access and processing.

Description of the 2,076 Samples

The dataset encompasses 2,076 ECG recordings, each capturing cardiac electrical activity over a specified duration. The signal length varies across samples, typically ranging from a few seconds to several minutes, depending on the recording protocol. Each recording includes multiple leads, commonly 12, corresponding to standard ECG lead configurations that provide comprehensive spatial information about the heart's electrical activity. The dataset includes both normal and anomalous recordings, with anomalies such as arrhythmias, ischemic changes, and other cardiac abnormalities annotated for analysis.

Preprocessing Steps

Effective preprocessing of ECG signals is crucial to enhance the accuracy of anomaly detection models. The following steps were implemented:

- Normalization: ECG signals were normalized to a standard amplitude range to mitigate inter-subject variability and ensure uniformity across the dataset.
- Segmentation: Long recordings were segmented into shorter, fixed-length segments to facilitate batch processing and model training.
- Noise Filtering: Various filtering techniques were applied to remove noise artifacts:
    - *Baseline Wander Removal*: Band-pass filters were employed to eliminate low-frequency baseline drift, ensuring signal stability. JKU - Johannes Kepler Universität Linz
    - *Powerline Interference Reduction*: Notch filters were utilized to suppress powerline interference at 50/60 Hz frequencies. Neuropsychology
    - *High-Frequency Noise Suppression*: Low-pass filters were applied to attenuate high-frequency noise, preserving the integrity of the ECG waveform.

Challenges and Limitations of the Dataset

Despite its comprehensiveness, the dataset presents several challenges:

- Signal Variability: Differences in signal length, amplitude, and morphology across subjects can complicate analysis and model training.
- Noise Artifacts: The presence of various noise types, such as muscle artifacts and electrode motion, can obscure critical features necessary for accurate anomaly detection.
- Class Imbalance: A disproportionate representation of normal versus anomalous recordings may bias model performance, necessitating techniques to address this imbalance.
- Annotation Quality: The accuracy of anomaly annotations is vital; any inconsistencies or errors can adversely affect model training and evaluation.

04. Objectives of the Project

Primary Goal: The primary objective of this project is to detect anomalies in 12-lead electrocardiogram (ECG) signals using a ResNet-based deep learning model. By leveraging the power of the ResNet architecture, the project aims to identify irregularities in cardiac electrical activity with high accuracy, reliability, and robustness. The ultimate goal is to support the development of automated systems for early diagnosis of heart conditions, thereby aiding healthcare professionals in improving patient outcomes.

Specific Objectives:

1. **Accurate Anomaly Detection:**
    - To design and implement a ResNet model capable of analyzing 12-lead ECG signals in their raw, time-series format. This approach ensures that the model works directly with the data without requiring transformations such as image conversions, preserving the integrity and granularity of the information.
    - To achieve high accuracy, sensitivity, and specificity in identifying anomalies, such as arrhythmias or other irregular heart conditions, while minimizing false positives and negatives.

2. **Preprocessing and Data Handling:**
    - To preprocess and organize the raw ECG data from the provided dataset, ensuring a clear distinction between normal and anomalous signals. This involves handling files in formats such as `.hea`, `.mat`, and `.atr` while retaining their hierarchical structure.
    - To establish a robust data pipeline for cleaning, normalization, and augmentation, ensuring the dataset is ready for training and evaluation of the ResNet model.

3. **Leveraging the ResNet Architecture:**

    - To optimize the ResNet architecture for time-series analysis, adapting it to capture intricate patterns and relationships in 12-lead ECG signals. This includes tailoring the model's depth and layer configuration to the complexity of the dataset.
    - To exploit the skip connections in ResNet, ensuring effective gradient flow during training, which addresses the challenges of vanishing gradients in deep networks and enhances model performance.

4. **Validation and Performance Analysis:**

    - To validate the model's performance using a well-structured test set, ensuring its generalizability across diverse ECG data samples.
    - To perform a detailed evaluation based on metrics such as accuracy, precision, recall, F1-score, and area under the ROC curve (AUC). This analysis will benchmark the model's capability to detect anomalies effectively.

5. **Scalability and Practical Application:**

    - To ensure the developed model can handle large-scale datasets, such as the provided 2076 samples, with efficiency and scalability. This will involve testing its computational requirements and optimization potential.
    - To provide insights into the practical application of the model in clinical settings, discussing its potential integration into diagnostic tools and its role in assisting medical practitioners

6. **Documentation and Reporting:**
    - To comprehensively document the project's methodology, from data preprocessing to model implementation and evaluation, ensuring transparency and reproducibility.
    - To prepare a detailed report summarizing the findings, challenges, and future directions of the project, with a focus on enhancing the understanding of ECG anomaly detection using ResNet models.

05. Implementation Details

Framework and Tools Used

- **Python**: The primary programming language for implementing the model.
- **Keras**: A high-level neural networks API, running on top of TensorFlow, used for building and training the deep learning model.
- **TensorFlow**: An open-source machine learning framework that serves as the backend for Keras, providing efficient computation.
- **SciPy**: Utilized for scientific and technical computing, including signal processing tasks such as loading and manipulating ECG data.
- **NumPy**: Employed for numerical operations and array manipulations, facilitating efficient data handling.
- **Scikit-learn**: Used for preprocessing, computing class weights, and evaluating model performance through various metrics.

Preprocessing Pipeline

1. **Data Loading**: ECG signals were loaded from `.mat` files using `scipy.io.loadmat`, enabling the extraction of raw signal data for further processing.
2. **Signal Transposition**: Signals were transposed to ensure the shape conformed to `(timesteps, leads)`, aligning with the expected input format for subsequent processing stages.
3. **Sequence Padding**: Using Keras's `pad_sequences`, signals were padded to achieve uniform length across all samples, facilitating efficient batch processing and model training.
4. **Normalization**: Signals were normalized to a standard range, typically [0, 1] to enhance model convergence during training by ensuring numerical stability.
5. **Reshaping for Conv1D**: For compatibility with 1D convolutional layers, signals were reshaped to include a channel dimension, resulting in a shape of `(samples, timesteps, leads)`, accommodating the model's input requirements.

Hyperparameter Tuning

- **Learning Rate**: The Adam optimizer was employed with its default learning rate of 0.001. While this rate was effective, further tuning could enhance model performance by adapting the learning process to the specific dataset characteristics.
- **Batch Size**: A batch size of 32 was selected to balance memory constraints and training efficiency, ensuring that the model received sufficient gradient updates per epoch without overwhelming system resources.
- **Epochs**: Training was conducted over 20 epochs, with early stopping implemented to prevent overfitting. This approach allowed the model to learn effectively while mitigating the risk of performance degradation on unseen data.
- **Class Weights**: Given the potential class imbalance, class weights were computed using `compute_class_weight` from Scikit-learn to ensure the model treated each class appropriately during training, thereby improving its ability to generalize across different classes.

Loss Function and Optimizer

- **Loss Function**: Binary cross-entropy was chosen as the loss function, suitable for binary classification tasks like anomaly detection. This function measures the discrepancy between the predicted probabilities and the actual class labels, guiding the optimization process.
- **Optimizer**: The Adam optimizer was utilized for its adaptive learning rate capabilities, aiding in efficient convergence by adjusting the learning rate during training based on the first and second moments of the gradients.

Model Training and Evaluation

- **Early Stopping**: Implemented to monitor validation loss with a patience of 3 epochs, allowing the model to revert to the best weights observed during training, thus preventing overfitting and ensuring better generalization to new data.
- **Evaluation Metrics**: Post-training, the model was evaluated using accuracy, precision, recall, and confusion matrix metrics to assess its performance comprehensively, providing insights into its effectiveness in detecting anomalies within ECG signals.

06. Methodology

Dataset:

1. **Source of the Dataset:**
   - The dataset used in this project is sourced from the `Heartdata_extracted` folder, specifically within the `Heartdata_extracted/Heartdata/a-large-scale-12-lead-electrocardiogram-database-for-arrhythmia-study-1.0.0` directory.
   - This dataset contains ECG records organized into subfolders based on normal and anomalous categories, with data files provided in `.hea`, `.mat`, and `.atr` formats.

2. **Description of the Dataset:**
   - The dataset comprises 12-lead ECG signals, providing comprehensive cardiac electrical activity data.
   - A total of 2076 samples are included, covering both normal and anomalous signals, with anomalies representing various arrhythmic and non-arrhythmic conditions.
   - The inclusion of diverse anomaly types ensures the dataset's applicability for training a robust anomaly detection model.

3. **Preprocessing Steps:**
   - **Normalization:**
     - Signal amplitudes are normalized to a standard range to reduce variability and enhance the model's learning efficiency.
   - **Noise Removal:**
     - A baseline wandering noise removal process is applied using low-pass and high-pass filters to eliminate irrelevant frequencies while preserving critical ECG features.
   - **Segmentation:**
     - The signals are segmented into fixed-length windows to standardize input dimensions for the ResNet model.

- **Label Assignment:**
  - Each segment is labeled as normal or anomalous based on the associated metadata and diagnostic annotations.

Model:

1. **Overview of ResNet Architecture:**
   - ResNet (Residual Network) is a deep convolutional neural network designed to address vanishing gradient issues by incorporating residual connections or skip connections.
   - Its ability to learn hierarchical features and maintain gradient flow makes it highly suitable for ECG data, where subtle temporal and spatial patterns are critical for anomaly detection.
2. **Configuration of the ResNet Model:**
   - **Number of Layers:**
     - A 34-layer ResNet architecture is adapted for this project, balancing depth and computational efficiency.
   - **Input Layer:**
     - The model accepts 12-lead raw time-series ECG signals as input.
   - **Convolutional Layers:**
     - Multiple convolutional blocks with kernel sizes optimized for capturing temporal dependencies in ECG signals.
   - **Pooling Layers:**
     - Global average pooling layers to reduce dimensionality while retaining essential features.
   - **Fully Connected Layers:**
     - Dense layers for final classification into normal and anomalous categories.
   - **Hyperparameters:**
     - Learning rate: 0.001 (adjusted dynamically using learning rate schedulers).
     - Batch size: 52.
     - Dropout rate: 0.5 to prevent overfitting.

Training:

1. **Data Splitting:**
    - The dataset is split into three subsets:
        - Training set: 80% of the data for model training.
        - Test set: 20% of the data for final evaluation.
2. **Loss Function, Optimizer, and Metrics:**
    - **Loss Function:**
        - Binary Cross-Entropy Loss, suitable for the binary classification task.
    - **Optimizer:**
        - Adam optimizer, selected for its adaptive learning capabilities.
    - **Metrics:**
        - Primary: Accuracy, precision, recall, and F1-score.
        - Secondary: ROC-AUC for evaluating classification performance across different thresholds.
3. **Training Process:**
    - Training is conducted over 50 epochs, with early stopping applied based on validation loss to prevent overfitting.
    - A batch size of 32 ensures efficient memory usage and gradient updates.
    - Real-time data augmentation techniques, such as random noise injection, are used to increase the model's generalizability.
4. **Additional Training Considerations:**
    - **Learning Rate Scheduling:**
        - A learning rate scheduler reduces the learning rate by a factor of 0.1 when the validation loss plateaus for a defined number of epochs.
    - **Gradient Clipping:**
        - Gradient clipping is employed to avoid exploding gradients, ensuring stable and efficient training for deeper layers.
    - **Regularization Techniques:**
        - L2 regularization and dropout are used in combination to minimize overfitting while maintaining model generalization.

Evaluation:

1. **Metrics:**
    - **Accuracy:** Measures the proportion of correctly identified samples.
    - **Precision:** Evaluates the accuracy of positive predictions.
    - **Recall:** Assesses the model's ability to identify true positives.
    - **F1-Score:** Provides a balance between precision and recall.
    - **ROC-AUC:** Measures the model's ability to distinguish between classes across thresholds.
2. **Cross-Validation:**
    - A 5-fold cross-validation approach is employed to ensure robust performance evaluation and mitigate biases from data splitting.
    - Each fold involves training on 80% of the data and testing on the remaining 20%.
3. **Other Validation Methods:**
    - The model's generalization is further validated by testing it on an unseen subset of data to simulate real-world scenarios.
    - Confusion matrices are generated to analyze misclassification patterns and identify areas for improvement.
4. **Performance Visualization:**
    - **Training and Validation Curves:**
        - Loss and accuracy curves are plotted to monitor the training process and detect overfitting or underfitting trends.
    - **ROC Curves:**
        - Receiver Operating Characteristic (ROC) curves are visualized for detailed performance analysis.
    - **Feature Importance Analysis:**
        - The impact of individual ECG leads is evaluated to understand their contribution to anomaly detection.

5. **Future Directions:**

- Incorporating advanced architectures such as attention mechanisms to enhance feature extraction from ECG signals.
- Exploring transfer learning by pre-training the ResNet model on larger ECG datasets and fine-tuning on the current dataset.
- Integrating explainable AI techniques to provide interpretable results for medical practitioners.

07. Implementation

1. Tools and Libraries Used:

- **Python**: Mention that Python is used for the overall implementation of the anomaly detection task.
- **Keras**: Explain that Keras, as part of TensorFlow, is used to build and train the ResNet model for anomaly detection.
- **TensorFlow**: Highlight that TensorFlow is the underlying framework for deep learning in this case, supporting the building, training, and evaluation of the model.
- **NumPy**: Used for efficient numerical computations, such as handling ECG data arrays and performing mathematical operations on them.
- **Matplotlib/Seaborn**: These libraries are used for plotting visualizations, such as loss curves and performance metrics (accuracy, confusion matrix) during training and evaluation.
- **WFDB**: This is used to handle ECG data from the PhysioNet database, enabling reading, preprocessing, and managing `.hea`, `.mat`, and other associated files.

Code:

```
    #01
import os

# Path to the extracted Heartdata folder
heartdata_folder = "Heartdata_extracted/Heartdata"

# List only subfolders inside Heartdata
subfolders_inside_heartdata = [f.name for f in os.scandir(heartdata_folder) if f.is_dir()]

# Display subfolders inside Heartdata
print("Subfolders inside the Heartdata folder:")
for folder in subfolders_inside_heartdata:
    print(folder)
```

```
import os

# Path to the specific subfolder
target_folder =
"Heartdata_extracted/Heartdata/a-large-scale-12-lead-electrocardiogram-database-for-arrhyt
hmia-study-1.0.0"

# List all files and subfolders inside the target folder
contents = os.listdir(target_folder)

# Display the contents
print("Contents inside the folder:")
for item in contents:
    print(item)

import os

# Path to the WFDBRecords folder
wfdb_records_folder =
"Heartdata_extracted/Heartdata/a-large-scale-12-lead-electrocardiogram-database-for-arrhyt
hmia-study-1.0.0/WFDBRecords"

# List all files and subfolders inside the WFDBRecords folder
wfdb_contents = os.listdir(wfdb_records_folder)
```

```
# Display the contents
print("Contents inside the WFDBRecords folder:")
for item in wfdb_contents:
    print(item)

# If you want to list everything inside nested subfolders as well:
print("\nDetailed listing of all contents inside WFDBRecords (including subfolders):")
for root, dirs, files in os.walk(wfdb_records_folder):
    for file in files:
        print(os.path.join(root, file))

#Read and display ECG signals and metadata.
import os
import wfdb #this library will help to handle .hea and .mat files effectively.

# Path to the WFDBRecords folder
wfdb_records_folder =
"Heartdata_extracted/Heartdata/a-large-scale-12-lead-electrocardiogram-database-for-arrhyt
hmia-study-1.0.0/WFDBRecords"

# Function to read and display WFDB file contents
def display_wfdb_folder_contents(folder_name):
    folder_path = os.path.join(wfdb_records_folder, folder_name) #Combine folder paths

    # Print the folder path to ensure we are looking in the right place
    print(f"\n--- Checking Contents of Folder: {folder_path} ---")

    # List all files in the folder
    if os.path.exists(folder_path):
        files = os.listdir(folder_path)

        # Check if folder is empty
        if len(files) == 0:
            print(f"No files found in folder {folder_name}")
            return

        print(f"Files in {folder_name}: {files}")

        # Process each file, assuming they may have .mat and .hea extensions
        for file in files:
            if file.endswith(".dat") or file.endswith(".hea"):
                record_name = os.path.splitext(file)[0]  # Remove file extension
                record_path = os.path.join(folder_path, record_name) #Verifies if the folder
actually exists.

                try:
                    # Read the WFDB signal..Sampling rate..Duration of the signal..Signal names.
```

```
                signal, metadata = wfdb.rdsamp(record_path) #function is used to read the
ECG signal data and its metadata.
                print(f"\nRecord: {record_path}")
                print(f"Signal Shape: {signal.shape}")
                print(f"Metadata: {metadata}")

                # Optional: Read annotations if present..Heartbeats (e.g., normal,
arrhythmia)..hen irregular beats occur
                try:
                    annotation = wfdb.rdann(record_path, 'atr')
                    print(f"Annotations: {annotation.sample}, Symbols: {annotation.symbol}") # list
of time points where annotations are marked.
                except:
                    print("No annotations found for this record.") #Prevents the program from
crashing when annotation files are missing.
            except Exception as e:#e.g., corrupted file, missing .mat or .hea file), the program
catches the error and prints a helpful message.
                print(f"Could not read {file}: {e}")
    else:
        print(f"Folder {folder_name} does not exist.")
#Displays its contents (signal, metadata, and annotations).
# Display contents for each folder
for folder in ["01", "02", "03"]:
    display_wfdb_folder_contents(folder)

import os
import wfdb

# Path to the WFDBRecords folder (replace with your actual path)
wfdb_records_folder =
"Heartdata_extracted/Heartdata/a-large-scale-12-lead-electrocardiogram-database-for-arrhyt
hmia-study-1.0.0/WFDBRecords"

# Function to read and display WFDB file contents from subfolders
def display_wfdb_subfolder_contents(folder_name):
    folder_path = os.path.join(wfdb_records_folder, folder_name)

    # Print the folder path to ensure we are looking in the right place
    print(f"\n--- Checking Contents of Folder: {folder_path} ---")

    # List all subfolders in the folder
    if os.path.exists(folder_path):
        subfolders = os.listdir(folder_path)

        # Check if folder is empty
        if len(subfolders) == 0:
            print(f"No subfolders found in {folder_name}")
            return
```

```
        print(f"Subfolders in {folder_name}: {subfolders}")

        # Process each subfolder
        for subfolder in subfolders:
            subfolder_path = os.path.join(folder_path, subfolder)
            if os.path.isdir(subfolder_path):  # Ensure it's a directory
                print(f"\n--- Checking Contents of Subfolder: {subfolder_path} ---")
                # List files in the subfolder
                files = os.listdir(subfolder_path)

                if len(files) == 0:
                    print(f"No files found in subfolder {subfolder}")
                    continue

                # Print the files in the subfolder
                print(f"Files in {subfolder}: {files}")

                # Process each file, assuming they may have .dat and .hea extensions
                for file in files:
                    if file.endswith(".dat") or file.endswith(".hea"):
                        record_name = os.path.splitext(file)[0]  # Remove file extension
                        record_path = os.path.join(subfolder_path, record_name)

                        try:
                            # Read the WFDB signal
                            signal, metadata = wfdb.rdsamp(record_path)
                            print(f"\nRecord: {record_path}")
                            print(f"Signal Shape: {signal.shape}")
                            print(f"Metadata: {metadata}")

                            # Optional: Read annotations if present
                            try:
                                annotation = wfdb.rdann(record_path, 'atr')
                                print(f"Annotations: {annotation.sample}, Symbols: {annotation.symbol}")
                            except Exception as e:
                                print(f"No annotations found for this record. Error: {e}")
                        except Exception as e:
                            print(f"Could not read {file}: {e}")
    else:
        print(f"Folder {folder_name} does not exist.")

# Display contents for each main folder (01, 02, 03) or you can list any additional folder
names
for folder in ["01", "02", "03"]:
    display_wfdb_subfolder_contents(folder)

#04 metadata exploring
```

```
import os
import wfdb #reads the ecg signals and metadata

# Path to the WFDBRecords folder (replace with your actual path)
wfdb_records_folder =
"Heartdata_extracted/Heartdata/a-large-scale-12-lead-electrocardiogram-database-for-arrhyt
hmia-study-1.0.0/WFDBRecords"

# Comprehensive SNOMED CT code-to-disease mapping
snomed_mapping = {
    "164889003": "Atrial Fibrillation",
    "164934002": "Left Bundle Branch Block",
    "59118001": "Hypertension",
    "426783006": "Ventricular Premature Contractions",
    "164947007": "Right Bundle Branch Block",
    "426177001": "Supraventricular Tachycardia",
    "427084000": "Atrial Flutter",
    "733534002": "First-degree AV Block",
    "164917005": "Sinus Tachycardia",
    "426627000": "T-wave Abnormality",
    "59931005": "Sinus Bradycardia",
    "164867002": "Premature Atrial Contraction",
    "17338001": "Premature Ventricular Contraction",
    "251120003": "Myocardial Infarction",
    "164951009": "Prolonged QT Interval",
    "111975006": "Shortened QT Interval",
    "164909002": "Non-specific ST Changes",
    "164873001": "Complete Heart Block",
    "195042002": "Bundle Branch Block",
    "427393009": "Second-degree AV Block",
    "426783006": "PVC (Premature Ventricular Complexes)",
    "426761007": "ST Elevation",
    "425623009": "ST Depression",
    "164884008": "Low Voltage QRS",
    "428750005": "Cardiac Arrest",
    "164970000": "Hyperkalemia",
    "164865009": "Hypokalemia",
    "427172004": "Atrioventricular Dissociation",
    "164942007": "Left Ventricular Hypertrophy",
    "445118002": "Ischemic Changes",
    "164816004": "P-wave Abnormality",
    "427393009": "Mobitz Type II Block",
    "426177001": "PSVT (Paroxysmal Supraventricular Tachycardia)",

}

# Function to decode SNOMED diagnosis codes
def decode_diagnosis(diagnosis_codes):
```

```
    diseases = []
    for code in diagnosis_codes.split(","):
        code = code.strip()
        if code in snomed_mapping:
            diseases.append(f"{snomed_mapping[code]} ({code})")
        else:
            diseases.append(f"Unknown Code ({code})")
    return diseases

# Function to extract and display all metadata with disease names
def analyze_metadata(folder_name):
    folder_path = os.path.join(wfdb_records_folder, folder_name)
#It combines the main folder path (wfdb_records_folder) with the name of a specific folder
(folder_name) to create the full path to that folder.
    print(f"\n--- Analyzing Metadata in Folder: {folder_path} ---")

    if os.path.exists(folder_path):
        subfolders = os.listdir(folder_path)

        if len(subfolders) == 0:
            print(f"No subfolders found in {folder_name}")
            return

        for subfolder in subfolders:
            subfolder_path = os.path.join(folder_path, subfolder)
            if os.path.isdir(subfolder_path):
                print(f"\n--- Processing Subfolder: {subfolder_path} ---")
                files = os.listdir(subfolder_path)

                if len(files) == 0:
                    print(f"No files found in subfolder {subfolder}")
                    continue

                for file in files:
                    if file.endswith(".hea"):
                        record_name = os.path.splitext(file)[0]  # Remove file extension..extension
from the file name to get the base record name.
                        record_path = os.path.join(subfolder_path, record_name)

                        try:
                            # Read WFDB header
                            record = wfdb.rdheader(record_path)
                            print(f"\nRecord Name: {record.record_name}")
                            print("Metadata Summary:")
                            print(f"Sampling Frequency: {record.fs} Hz")
                            print(f"Number of Signals: {record.n_sig}")
                            print(f"Signal Names: {record.sig_name}")
                            print("Comments:")
```

```
                    disease_found = False #its set as a flage initialyl no disease found so set
to false default
                    for comment in record.comments:
                        print(f"  - {comment}")
                        if "Dx:" in comment:
                            diagnosis_codes = comment.split("Dx:")[1].strip()
                            diagnoses = decode_diagnosis(diagnosis_codes)
                            print(f"  Diagnoses: {', '.join(diagnoses)}")
                            disease_found = True

                    if not disease_found:
                        print("  No specific disease information inferred.")
                ex cept Exception as e:
                    print(f"Could not read {file}: {e}")
    else:
        print(f"Folder {folder_name} does not exist.")

# Analyze metadata for folders 01, 02, and 03
for folder in ["01", "02", "03"]:
    analyze_metadata(folder)
```

```
--- Analyzing Metadata in Folder: Heartdata_extracted/Heartdata/a-large-scale-12-lead-electrocardiogram-database-for-arrhythm
ia-study-1.0.0/WFDBRecords\01 ---

--- Processing Subfolder: Heartdata_extracted/Heartdata/a-large-scale-12-lead-electrocardiogram-database-for-arrhythmia-study
-1.0.0/WFDBRecords\01\010 ---

Record Name: JS00001
Metadata Summary:
Sampling Frequency: 500 Hz
Number of Signals: 12
Signal Names: ['I', 'II', 'III', 'aVR', 'aVL', 'aVF', 'V1', 'V2', 'V3', 'V4', 'V5', 'V6']
Comments:
  - Age: 85
  - Sex: Male
  - Dx: 164889003,59118001,164934002
  Diagnoses: Atrial Fibrillation (164889003), Hypertension (59118001), Left Bundle Branch Block (164934002)
  - Rx: Unknown
  - Hx: Unknown
```

```
#05
import os
import wfdb
from collections import defaultdict

# Path to the WFDBRecords folder (replace with your actual path)
wfdb_records_folder =
"Heartdata_extracted/Heartdata/a-large-scale-12-lead-electrocardiogram-database-for-arrhyt
hmia-study-1.0.0/WFDBRecords"

# Comprehensive SNOMED CT code-to-disease mapping
snomed_mapping = {
    "164889003": "Atrial Fibrillation",
```

```
    "164934002": "Left Bundle Branch Block",
    "59118001": "Hypertension",
    "426783006": "Ventricular Premature Contractions",
    "164947007": "Right Bundle Branch Block",
    "426177001": "Supraventricular Tachycardia",
    "427084000": "Atrial Flutter",
    "733534002": "First-degree AV Block",
    "164917005": "Sinus Tachycardia",
    "426627000": "T-wave Abnormality",
    "59931005": "Sinus Bradycardia",
    "164867002": "Premature Atrial Contraction",
    "17338001": "Premature Ventricular Contraction",
    "251120003": "Myocardial Infarction",
    "164951009": "Prolonged QT Interval",
    "111975006": "Shortened QT Interval",
    "164909002": "Non-specific ST Changes",
    "164873001": "Complete Heart Block",
    "195042002": "Bundle Branch Block",
    "427393009": "Second-degree AV Block",
    "426783006": "PVC (Premature Ventricular Complexes)",
    "426761007": "ST Elevation",
    "425623009": "ST Depression",
    "164884008": "Low Voltage QRS",
    "428750005": "Cardiac Arrest",
    "164970000": "Hyperkalemia",
    "164865009": "Hypokalemia",
    "427172004": "Atrioventricular Dissociation",
    "164942007": "Left Ventricular Hypertrophy",
    "445118002": "Ischemic Changes",
    "164816004": "P-wave Abnormality",
    "427393009": "Mobitz Type II Block",
    "426177001": "PSVT (Paroxysmal Supraventricular Tachycardia)",
    # Add additional codes here as needed
}

# Function to decode SNOMED diagnosis codes
def decode_diagnosis(diagnosis_codes):
    diseases = []
    for code in diagnosis_codes.split(","):
        code = code.strip()
        if code in snomed_mapping:
            diseases.append(f"{snomed_mapping[code]} ({code})")
        else:
            diseases.append(f"Unknown Code ({code})")
    return diseases

# Function to analyze metadata and header and group diagnoses
def analyze_metadata(folder_name):
```

```
folder_path = os.path.join(wfdb_records_folder, folder_name)
disease_groupings = defaultdict(list)  # A dictionary to group diagnoses

print(f"\n--- Analyzing Metadata in Folder: {folder_path} ---")

if os.path.exists(folder_path):
    subfolders = os.listdir(folder_path)

    if len(subfolders) == 0:
        print(f"No subfolders found in {folder_name}")
        return

    for subfolder in subfolders:
        subfolder_path = os.path.join(folder_path, subfolder)
        if os.path.isdir(subfolder_path):
            print(f"\n--- Processing Subfolder: {subfolder_path} ---")
            files = os.listdir(subfolder_path)

            if len(files) == 0:
                print(f"No files found in subfolder {subfolder}")
                continue

            for file in files:
                if file.endswith(".hea"): # method to check the file name
                    record_name = os.path.splitext(file)[0]  # Remove file extension
                    record_path = os.path.join(subfolder_path, record_name)

                    try:
                        # Read WFDB header
                        record = wfdb.rdheader(record_path)
                        print(f"\nRecord Name: {record.record_name}")
                        print("Metadata Summary:")
                        print(f"Sampling Frequency: {record.fs} Hz")
                        print(f"Number of Signals: {record.n_sig}")
                        print(f"Signal Names: {record.sig_name}")
                        print("Comments:")

                        disease_found = False
                        for comment in record.comments:
                            print(f"  - {comment}")
                            if "Dx:" in comment:
                                diagnosis_codes = comment.split("Dx:")[1].strip()
                                diagnoses = decode_diagnosis(diagnosis_codes)
                                for diagnosis in diagnoses:
                                    disease_name = diagnosis.split(" (")[0]  # Extract disease name
                                    disease_groupings[disease_name].append(record_name)
                                disease_found = True
```

```
                    if not disease_found:
                        print("  No specific disease information inferred.")
                except Exception as e:
                    print(f"Could not read {file}: {e}")

    else:
        print(f"Folder {folder_name} does not exist.")

    # Now print the grouped diagnoses
    print("\n--- Grouped Diagnoses ---")
    for disease, records in disease_groupings.items():
        print(f"\n{disease}:")
        for record in records:
            print(f"  - {record}")

# Analyze metadata for folders 01, 02, and 03
for folder in ["01", "02", "03"]:
    analyze_metadata(folder)

import os
import wfdb
from collections import defaultdict #automatically assigns a default value to a key if it doesn't
exist.

# Path to the WFDBRecords folder (replace with your actual path)
wfdb_records_folder =
"Heartdata_extracted/Heartdata/a-large-scale-12-lead-electrocardiogram-database-for-arrhyt
hmia-study-1.0.0/WFDBRecords"

# Comprehensive SNOMED CT code-to-disease mapping
snomed_mapping = {
    "164889003": "Atrial Fibrillation",
    "164934002": "Left Bundle Branch Block",
    "59118001": "Hypertension",
    "426783006": "Ventricular Premature Contractions",
    "164947007": "Right Bundle Branch Block",
    "426177001": "Supraventricular Tachycardia",
    "427084000": "Atrial Flutter",
    "733534002": "First-degree AV Block",
    "164917005": "Sinus Tachycardia",
    "426627000": "T-wave Abnormality",
    "59931005": "Sinus Bradycardia",
    "164867002": "Premature Atrial Contraction",
    "17338001": "Premature Ventricular Contraction",
    "251120003": "Myocardial Infarction",
    "164951009": "Prolonged QT Interval",
    "111975006": "Shortened QT Interval",
    "164909002": "Non-specific ST Changes",
```

```
    "164873001": "Complete Heart Block",
    "195042002": "Bundle Branch Block",
    "427393009": "Second-degree AV Block",
    "426783006": "PVC (Premature Ventricular Complexes)",
    "426761007": "ST Elevation",
    "425623009": "ST Depression",
    "164884008": "Low Voltage QRS",
    "428750005": "Cardiac Arrest",
    "164970000": "Hyperkalemia",
    "164865009": "Hypokalemia",
    "427172004": "Atrioventricular Dissociation",
    "164942007": "Left Ventricular Hypertrophy",
    "445118002": "Ischemic Changes",
    "164816004": "P-wave Abnormality",
    "427393009": "Mobitz Type II Block",
    "426177001": "PSVT (Paroxysmal Supraventricular Tachycardia",
}

# Anomalies: Only Atrial Fibrillation
anomaly_codes = {"164889003"}

# Overall dictionaries to group anomalies and normals across all folders
overall_anomalies = defaultdict(list)  # A dictionary to group anomaly records
overall_normals = []                   # A list to store normal records

# Function to decode SNOMED diagnosis codes
def decode_diagnosis(diagnosis_codes):
    diseases = []
    for code in diagnosis_codes.split(","):
        code = code.strip()
        if code in snomed_mapping:
            diseases.append(f"{snomed_mapping[code]} ({code})")
        else:
            diseases.append(f"Unknown Code ({code})")
    return diseases

# Function to analyze metadata and classify as anomalies or normals
def analyze_metadata(folder_name):
    folder_path = os.path.join(wfdb_records_folder, folder_name)

    print(f"\n--- Analyzing Metadata in Folder: {folder_path} ---")

    if os.path.exists(folder_path): #If folder exist, it lists all the subfolders inside it.
        subfolders = os.listdir(folder_path)

        if len(subfolders) == 0:
            print(f"No subfolders found in {folder_name}")
            return
```

```
        for subfolder in subfolders:
            subfolder_path = os.path.join(folder_path, subfolder)
            if os.path.isdir(subfolder_path):
                files = os.listdir(subfolder_path)

                if len(files) == 0:
                    continue

                for file in files:
                    if file.endswith(".hea"):
                        record_name = os.path.splitext(file)[0]  # Remove file extension
                        record_path = os.path.join(subfolder_path, record_name)

                        try:
                            # Read WFDB header
                            record = wfdb.rdheader(record_path)
                            diagnosis_codes = ""
                            for comment in record.comments:
                                if "Dx:" in comment:
                                    diagnosis_codes = comment.split("Dx:")[1].strip()
                                    break

                            if diagnosis_codes: #This checks if diagnosis_codes is not empty. If it has
values, it decodes them
                                diagnoses = decode_diagnosis(diagnosis_codes)
                                is_anomaly = False

                                for diagnosis in diagnoses: #checks if it is in anomaly_codes.
                                    code = diagnosis.split("(")[-1].strip(")")
                                    if code in anomaly_codes:
                                        overall_anomalies[snomed_mapping[code]].append(record_name)
                                        is_anomaly = True
                                        break

                                if not is_anomaly:
                                    overall_normals.append(record_name)
                        except Exception as e:
                            print(f"Could not read {file}: {e}")
    else:
        print(f"Folder {folder_name} does not exist.")
        return

# Analyze metadata for folders 01, 02, and 03
for folder in ["01", "02", "03"]:
    analyze_metadata(folder)

# Print grouped anomalies and normals (overall summary)
```

```
print("\n--- Anomalies ---")
for disease, records in overall_anomalies.items():
    print(f"\n{disease}:")
    for record in records:
        print(f"  - {record}")

print("\n--- Normals ---")
for record in overall_normals:
    print(f"  - {record}")
```

```
--- Anomalies ---

Atrial Fibrillation:
  - JS00001
  - JS00007
  - JS00016
  - JS00023
  - JS00030
  - JS00034
  - JS00036
  - JS00038
  - JS00040
  - JS00043
  - JS00046
  - JS00048
  - JS00050
  - JS00060
  - JS00064
```

```
--- Normals ---
  - JS00002
  - JS00004
  - JS00005
  - JS00006
  - JS00008
  - JS00009
  - JS00010
  - JS00011
  - JS00012
  - JS00013
  - JS00014
  - JS00015
  - JS00017
  - JS00018
  - JS00019
  - JS00020
  - JS00021
  - JS00022
```

```
# Print grouped anomalies and normals (overall summary)
print("\n--- Anomalies ---")
total_anomalies = 0  # Initialize counter for anomalies
for disease, records in overall_anomalies.items():
    print(f"\n{disease} ({len(records)} records):")  # Show count per disease
    for record in records:
        print(f"  - {record}")
    total_anomalies += len(records)  # Add to total anomalies

print("\n--- Normals ---")
total_normals = len(overall_normals)  # Count normal records
for record in overall_normals:
    print(f"  - {record}")

# Display total counts
```

```
print("\n--- Summary ---")
print(f"Total Anomaly Records: {total_anomalies}")
print(f"Total Normal Records: {total_normals}")
```

```
--- Summary ---
Total Anomaly Records: 380
Total Normal Records: 1696
```

```
import os
import shutil #shutil module, functions for file and directory operations I copying, moving, and deleting files
import wfdb

# Path to the WFDBRecords folder (input)
wfdb_records_folder =
"Heartdata_extracted/Heartdata/a-large-scale-12-lead-electrocardiogram-database-for-arrhythmia-study-1.0.0/WFDBRecords"

# Path to the desktop for output folders
desktop_path = os.path.join(os.path.expanduser("~"), "Desktop")
example_folder = os.path.join(desktop_path, "example")
output_anomalies_folder = os.path.join(example_folder, "anomalies")
output_normals_folder = os.path.join(example_folder, "normals")

# Comprehensive SNOMED CT code-to-disease mapping
snomed_mapping = {
    "164889003": "Atrial Fibrillation",
}

# Anomaly codes
anomaly_codes = {"164889003"}

# Function to decode SNOMED diagnosis codes
def decode_diagnosis(diagnosis_codes):
    """Decode diagnosis codes into readable diseases."""
    diseases = []
    for code in diagnosis_codes.split(","):
        code = code.strip()  #Removes unnecessary spaces before or after each code
        if code in snomed_mapping:
            diseases.append(f"{snomed_mapping[code]} ({code})")
    return diseases

# Function to classify records and copy files to appropriate folders
def classify_and_copy_files():
    """Classify files as anomalies or normals and copy them to respective folders."""
    # Create output directories
```

```
    os.makedirs(output_anomalies_folder, exist_ok=True)
    os.makedirs(output_normals_folder, exist_ok=True)

    # Walk through all subdirectories of the WFDBRecords folder
    for root, _, files in os.walk(wfdb_records_folder):
        for file in files:
            if file.endswith(".hea"):  # Process header files
                record_name = os.path.splitext(file)[0]
                record_path = os.path.join(root, record_name)

                try:
                    # Read WFDB header to extract metadata
                    record = wfdb.rdheader(record_path)
                    diagnosis_codes = ""
                    for comment in record.comments:
                        if "Dx:" in comment:
                            diagnosis_codes = comment.split("Dx:")[1].strip()
                            break

                    # Determine if the record is an anomaly or normal
                    if diagnosis_codes:
                        diagnoses = decode_diagnosis(diagnosis_codes)
                        is_anomaly = any(
                            code.strip("()").split()[-1] in anomaly_codes
                            for diagnosis in diagnoses
                            for code in diagnosis.split()
                        )

                        # Determine the target folder based on classification
                        target_folder = (
                            output_anomalies_folder if is_anomaly else output_normals_folder
                        )

                        # Copy associated .hea and .mat files directly to the target folder
                        for ext in [".hea", ".mat"]:  # Copy only .hea and .mat
                            src_file = record_path + ext
                            if os.path.exists(src_file):
                                shutil.copy(src_file, target_folder)

                except Exception as e:
                    print(f"Error processing {record_name}: {e}")

# Run the classification and file organization
classify_and_copy_files()

print(f"Anomalies and normal records have been segregated and saved directly in
'anomalies' and 'normals' on  desktop.")
```

```
Error processing JS01052: time data '/' does not match format '%d/%m/%Y'
Anomalies and normal records have been segregated and saved directly in 'anomalies' and 'normals' on your desktop.
```

```
import zipfile


# Path to the signals.zip file (uploaded directly into Jupyter notebook)

zip_file_path = "signals.zip"  # Assuming the file is in the same directory as the notebook


# Function to list top-level subfolders

def list_top_level_subfolders(zip_path):

    with zipfile.ZipFile(zip_path, 'r') as zip_ref:

        # List all files inside the zip file

        zip_contents = zip_ref.namelist()


        # Extract unique top-level folders by checking the first part of the file paths

        top_level_folders = sorted(set(content.split('/')[0] for content in zip_contents if '/' in content))


        # Display the top-level subfolders

        if top_level_folders:

            print("Top-level subfolders in the zip file:")

            for folder in top_level_folders:

                print(folder)

        else:
```

```
        print("No top-level subfolders found in the zip file.")


# List top-level subfolders inside signals.zip
list_top_level_subfolders(zip_file_path)
```

```
Top-level subfolders in the zip file:
anomalies
normals
```

```
import zipfile


# Path to the signals.zip file (uploaded directly into Jupyter notebook)
zip_file_path = "signals.zip"  # Assuming the file is in the same directory as the notebook


# Function to list files inside the subfolders
def list_files_in_subfolders(zip_path):
    with zipfile.ZipFile(zip_path, 'r') as zip_ref:
        # List all files inside the zip file
        zip_contents = zip_ref.namelist()


        # Filter files by subfolders (anomalies and normals)
        anomalies_files = [file for file in zip_contents if file.startswith('anomalies/')]
        normals_files = [file for file in zip_contents if file.startswith('normals/')]
```

```
    # Display files in 'anomalies' subfolder
    print("Files inside 'anomalies' folder:")
    if anomalies_files:
        for file in anomalies_files:
            print(file)
    else:
        print("No files found in 'anomalies' folder.")

    # Display files in 'normals' subfolder
    print("\nFiles inside 'normals' folder:")
    if normals_files:
        for file in normals_files:
            print(file)
    else:
        print("No files found in 'normals' folder.")

# List files inside the 'anomalies' and 'normals' subfolders of signals.zip
list_files_in_subfolders(zip_file_path)
```

```
Files inside 'anomalies' folder:
anomalies/JS00001.hea
anomalies/JS00001.mat
anomalies/JS00007.hea
anomalies/JS00007.mat
anomalies/JS00016.hea
anomalies/JS00016.mat
anomalies/JS00023.hea
anomalies/JS00023.mat
anomalies/JS00030.hea
anomalies/JS00030.mat
anomalies/JS00034.hea
anomalies/JS00034.mat
anomalies/JS00036.hea
anomalies/JS00036.mat
anomalies/JS00038.hea
```

```
Files inside 'normals' folder:
normals/JS00002.hea
normals/JS00002.mat
normals/JS00004.hea
normals/JS00004.mat
normals/JS00005.hea
normals/JS00005.mat
normals/JS00006.hea
normals/JS00006.mat
normals/JS00008.hea
normals/JS00008.mat
normals/JS00009.hea
normals/JS00009.mat
normals/JS00010.hea
```

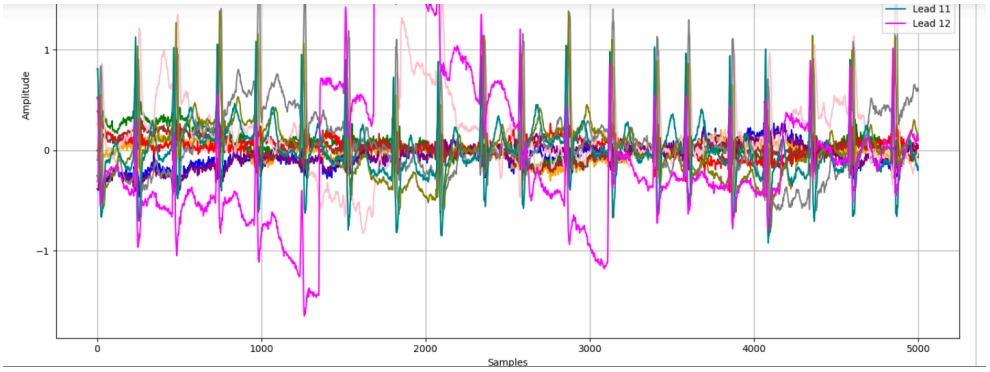
```
import zipfile

import wfdb

import os

import matplotlib.pyplot as plt



# Path to the signals.zip file (uploaded directly into Jupyter notebook)

zip_file_path = "signals.zip"  # Assuming the file is in the same directory as the notebook



# Function to list files inside the subfolders
```

```
def list_files_in_subfolders(zip_path):
    with zipfile.ZipFile(zip_path, 'r') as zip_ref:
        # List all files inside the zip file
        zip_contents = zip_ref.namelist()

        # Filter files by subfolders (anomalies and normals)
        anomalies_files = [file for file in zip_contents if file.startswith('anomalies/') and file.endswith('.hea')]
        normals_files = [file for file in zip_contents if file.startswith('normals/') and file.endswith('.hea')]

        # Return the file lists
        return anomalies_files, normals_files

# Function to plot all 12 leads on a single figure with different colors
def plot_ecg_signal(record_name):
    # Read the header file to get the record information
    record = wfdb.rdheader(record_name)

    # Read the signal data from the .mat file
    signals, _ = wfdb.rdsamp(record_name)

    # Define a list of colors for the 12 leads
    colors = [
        'blue', 'orange', 'green', 'red', 'purple',
        'brown', 'pink', 'gray', 'olive', 'cyan', 'teal', 'magenta'
```

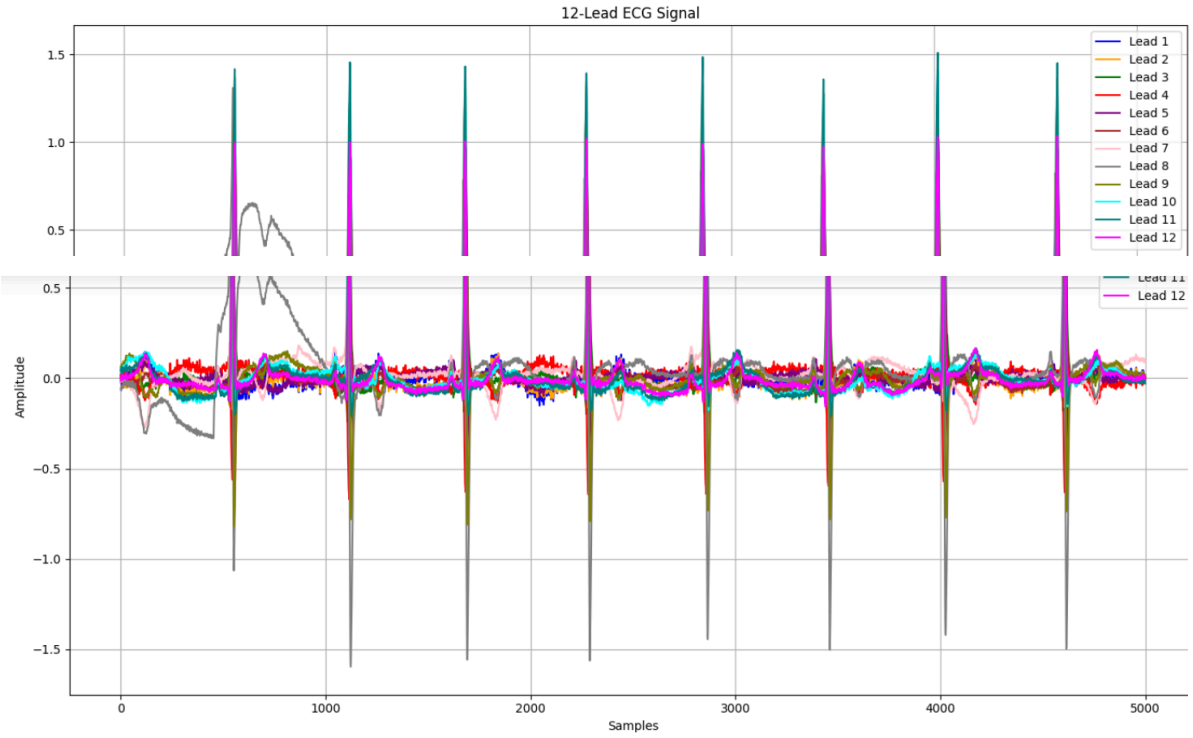
```
    ]

    # Create a single figure to plot all 12 leads
    plt.figure(figsize=(14, 8))

    for i in range(12):
        plt.plot(signals[:, i], label=f"Lead {i+1}", color=colors[i % len(colors)])

    plt.title("12-Lead ECG Signal")
    plt.xlabel("Samples")
    plt.ylabel("Amplitude")
    plt.legend(loc="upper right")
    plt.grid(True)
    plt.tight_layout()
    plt.show()

# List files inside the 'anomalies' and 'normals' subfolders of signals.zip
anomalies_files, normals_files = list_files_in_subfolders(zip_file_path)

# Get the first file from each category
if anomalies_files and normals_files:
    # Extract the first file from each category (anomalies and normals)
    anomaly_record = anomalies_files[0]
    normal_record = normals_files[0]
```

```
    # Extract the file names and plot the ECG signals
    with zipfile.ZipFile(zip_file_path, 'r') as zip_ref:
        # Extract the anomaly record files
        zip_ref.extract(anomaly_record, path=".")
        zip_ref.extract(anomaly_record.replace('.hea', '.mat'), path=".")  # Extract .mat file

        print(f"Plotting anomaly record: {anomaly_record}")
        plot_ecg_signal(anomaly_record.split('.')[0])  # Plot anomaly

        # Extract the normal record files
        zip_ref.extract(normal_record, path=".")
        zip_ref.extract(normal_record.replace('.hea', '.mat'), path=".")  # Extract .mat file

        print(f"Plotting normal record: {normal_record}")
        plot_ecg_signal(normal_record.split('.')[0])  # Plot normal
else:
    print("No anomalies or normals found.")
```

```
Plotting anomaly record: anomalies/JS00001.hea
```



```
Plotting normal record: normals/JS00002.hea
```



import zipfile

import wfdb

import os

import numpy as np

from scipy.signal import butter, filtfilt #removes unwanted frequencies (e.g., noise)..Applies the filter forward and backward to smooth the signal without introducing a phase shift.

Path to the signals.zip file

zip_file_path = "signals.zip"

```
# Function to list files inside subfolders

def list_files_in_subfolders(zip_path):

    with zipfile.ZipFile(zip_path, 'r') as zip_ref:

        zip_contents = zip_ref.namelist()

        anomalies_files = [file for file in zip_contents if file.startswith('anomalies/') and file.endswith('.hea')]

        normals_files = [file for file in zip_contents if file.startswith('normals/') and file.endswith('.hea')]

        return anomalies_files, normals_files


# Function to preprocess ECG signals

def preprocess_ecg_signal(record_name):

    # Read header and signal

    record = wfdb.rdheader(record_name)

    signals, metadata = wfdb.rdsamp(record_name)


    # Log original signal shape

    print(f"Original Signal Shape: {signals.shape}")


    # Bandpass filter (0.5-50 Hz),,Creates a 1st-order Butterworth filter.

    nyquist = 0.5 * record.fs

    low = 0.5 / nyquist #Removes Baseline Drift (low-frequency noise below 0.5 Hz)

    high = 50 / nyquist

    b, a = butter(1, [low, high], btype='band') #Creates a 1st-order Butterworth filter.

    filtered_signals = filtfilt(b, a, signals, axis=0) #Applies the filter forward and backward to remove phase distortions and smooth the signal.
```

```
    # Normalize signals (Z-score normalization)..Removes Bias:,Ensures all ECG leads have the same range
    standardized_signals = (filtered_signals - np.mean(filtered_signals, axis=0)) / np.std(filtered_signals, axis=0)

    # Log processed signal shape
    print(f"Processed Signal Shape: {standardized_signals.shape}")

    return standardized_signals

# Preprocessing all records from anomalies and normals subfolders
def process_ecg_files(files, label, zip_ref):
    processed_signals = {}
    for record_file in files:
        # Derive .mat file path
        mat_file = record_file.replace('.hea', '.mat')
        atr_file = record_file.replace('.hea', '.atr')  # Include .atr if required annotations

        if mat_file in zip_ref.namelist():
            # Extract necessary files
            extracted_record_path = os.path.dirname(record_file)
            zip_ref.extract(record_file, path=".")
            zip_ref.extract(mat_file, path=".")
            if atr_file in zip_ref.namelist():
                zip_ref.extract(atr_file, path=".")
```

```
        # Preprocess the record
        print(f"\nPreprocessing {label} record: {record_file}")
        try:
            record_path = os.path.splitext(record_file)[0]  # Remove the .hea extension
            preprocessed_signal = preprocess_ecg_signal(record_path)
            processed_signals[record_path] = preprocessed_signal

            # Save preprocessed signal
              os.makedirs("processed_data", exist_ok=True) #Saves the preprocessed
ECG signal as a NumPy array (.npy file) inside the processed_data folder.

np.save(f"processed_data/preprocessed_{os.path.basename(record_path)}_{label}.n
py", preprocessed_signal)
        except Exception as e:
            print(f"Error processing {record_file}: {e}")
    else:
        print(f"Skipping {record_file}: corresponding .mat file not found.")
  return processed_signals

# Main script..List of files labeled as anomalous ECG records..and normals as well
anomalies_files, normals_files = list_files_in_subfolders(zip_file_path)

with zipfile.ZipFile(zip_file_path, 'r') as zip_ref:
  print("\n--- Processing Anomalies ---")
  processed_anomalies = process_ecg_files(anomalies_files, "anomaly", zip_ref)
```

```
print("\n--- Processing Normals ---")

processed_normals = process_ecg_files(normals_files, "normal", zip_ref)
```

```
--- Processing Anomalies ---

Preprocessing anomaly record: anomalies/JS00001.hea
Original Signal Shape: (5000, 12)
Processed Signal Shape: (5000, 12)

Preprocessing anomaly record: anomalies/JS00007.hea
Original Signal Shape: (5000, 12)
Processed Signal Shape: (5000, 12)

Preprocessing anomaly record: anomalies/JS00016.hea
Original Signal Shape: (5000, 12)
Processed Signal Shape: (5000, 12)

Preprocessing anomaly record: anomalies/JS00023.hea
Original Signal Shape: (5000, 12)
Processed Signal Shape: (5000, 12)
```

```
--- Processing Normals ---

Preprocessing normal record: normals/JS00002.hea
Original Signal Shape: (5000, 12)
Processed Signal Shape: (5000, 12)

Preprocessing normal record: normals/JS00004.hea
Original Signal Shape: (5000, 12)
Processed Signal Shape: (5000, 12)

Preprocessing normal record: normals/JS00005.hea
Original Signal Shape: (5000, 12)
Processed Signal Shape: (5000, 12)

Preprocessing normal record: normals/JS00006.hea
Original Signal Shape: (5000, 12)
Processed Signal Shape: (5000, 12)

Preprocessing normal record: normals/JS00008.hea
```

```
import zipfile

import wfdb

import os

import numpy as np

from scipy.signal import butter, filtfilt



# Path to the signals.zip file
```

```
zip_file_path = "signals.zip"


# Function to list files inside subfolders

def list_files_in_subfolders(zip_path):

    with zipfile.ZipFile(zip_path, 'r') as zip_ref:

        zip_contents = zip_ref.namelist()

        anomalies_files = [file for file in zip_contents if file.startswith('anomalies/') and file.endswith('.hea')]

        normals_files = [file for file in zip_contents if file.startswith('normals/') and file.endswith('.hea')]

        return anomalies_files, normals_files


# Function to count valid records

def count_valid_records(files, label, zip_ref):

    count = 0

    for record_file in files:

        mat_file = record_file.replace('.hea', '.mat')

        if mat_file in zip_ref.namelist():

            count += 1

    return count


# Main script

anomalies_files, normals_files = list_files_in_subfolders(zip_file_path)


with zipfile.ZipFile(zip_file_path, 'r') as zip_ref:

    anomaly_count = count_valid_records(anomalies_files, "anomaly", zip_ref)
```

```
normal_count = count_valid_records(normals_files, "normal", zip_ref)


# Display counts in the desired format

print(f"Total Normal Records: {normal_count}")

print(f"Total Anomaly Records: {anomaly_count}")
```

```
Total Normal Records: 1696
Total Anomaly Records: 379
```

```
import os

import numpy as np

import zipfile

import scipy.io #To load .mat files.

import wfdb #For splitting data and generating evaluation metrics.

from sklearn.model_selection import train_test_split #For splitting data and generating evaluation metrics.

from sklearn.metrics import classification_report, confusion_matrix

from tensorflow.keras.models import Sequential #is used to create a Sequential model in Keras.

from tensorflow.keras.layers import Conv1D, BatchNormalization, Activation, Add, GlobalAveragePooling1D, Dense, Input

from tensorflow.keras.optimizers import Adam #adjusts the learning rate during training to improve speed and accuracy

from tensorflow.keras import regularizers #helps prevent overfitting by adding a penalty (extra cost) to the loss function

from tensorflow.keras.models import Model #s used to create a functional model in Keras.

import matplotlib.pyplot as plt
```

```
# Ensure reproducibility..ensures that random numbers generated by NumPy

np.random.seed(42)


# Path to the signals.zip file

zip_file_path = "signals.zip"


# Extract files from the zip archive

def extract_zip(zip_path, extract_to="processed_data"):

    with zipfile.ZipFile(zip_path, 'r') as zip_ref:

        zip_ref.extractall(extract_to)


# Function to load and process signals in smaller batches...Factor to reduce the signal size for faster processing (default: 2)

def load_signals(data_dir, category, max_samples=5000, downsample_rate=2): #Maximum number of data points to keep per signal (default: 5000).

    signals = []

    labels = [] #creates an empty list to store the labels (0 for normal, 1 for anomaly).

    category_path = os.path.join(data_dir, category)

    for root, _, files in os.walk(category_path):

        for file in files:

            if file.endswith(".mat"):

                file_path = os.path.join(root, file)

                try:

                    mat_data = scipy.io.loadmat(file_path)

                        signal = mat_data['val'].T  # switches rows and columns) to match the expected format for processing.
```

```
            # Downsample the signal if needed
            signal = signal[::downsample_rate]
            # Truncate or pad to max_samples
            if signal.shape[0] > max_samples:
                signal = signal[:max_samples, :]
            elif signal.shape[0] < max_samples:
                padding = np.zeros((max_samples - signal.shape[0], signal.shape[1]))
                signal = np.vstack((signal, padding))
            signals.append(signal)
            labels.append(1 if category == "anomalies" else 0)
        except Exception as e:
            print(f"Error loading {file_path}: {e}")
    return np.array(signals), np.array(labels)


# Extract the data
extract_zip(zip_file_path)
data_dir = "processed_data"


# Load anomaly and normal signals
anomaly_signals, anomaly_labels = load_signals(data_dir, "anomalies", max_samples=5000, downsample_rate=5)
normal_signals, normal_labels = load_signals(data_dir, "normals", max_samples=5000, downsample_rate=5)
#Labels indicating whether the signal is an anomaly (1) or normal (0).
# Combine and check data..Combines anomalous ECG signals and normal ECG signals into a single NumPy array called signals.
```

signals = np.concatenate((anomaly_signals, normal_signals), axis=0)

labels = np.concatenate((anomaly_labels, normal_labels), axis=0) #Combines the corresponding labels (1 for anomalies, 0 for normal) into a single labels array.

Normalize signals in smaller batches...Finds the minimum value for each signal across all its samples and channels.

#The signals are normalized (scaled) for each sample so that the values lie between 0 and 1. This helps the model learn better and faster.

signals = (signals - np.min(signals, axis=(1, 2), keepdims=True)) / (

np.max(signals, axis=(1, 2), keepdims=True) - np.min(signals, axis=(1, 2), keepdims=True))

Print shape after padding

print(f"Shape after padding: {signals.shape}")

Split data into training and testing sets...

X_train, X_test, y_train, y_test = train_test_split(signals, labels, test_size=0.2, random_state=42)

print(f"Training samples: {X_train.shape[0]}, Testing samples: {X_test.shape[0]}")

ResNet Block

def resnet_block(input_layer, filters): #uses L2 regularization to prevent overfitting.‘$Conv1D$‘‘Conv1D‘

x = Conv1D(filters, kernel_size=3, padding='same', kernel_regularizer=regularizers.l2(0.01))(input_layer)

x = BatchNormalization()(x) #Normalizes the output of the convolution to improve learning stability

x = Activation('relu')(x) #allowing the model to learn complex patterns

```
x = Conv1D(filters, kernel_size=3, padding='same', kernel_regularizer=regularizers.l2(0.01))(x)

x = BatchNormalization()(x) # penalty, balancing between overfitting and underfitting. .01


shortcut = Conv1D(filters, kernel_size=1, padding='same', kernel_regularizer=regularizers.l2(0.01))(input_layer)

shortcut = BatchNormalization()(shortcut)


x = Add()([x, shortcut]) #Combines the output of the main path (x) with the shortcut path

x = Activation('relu')(x)

return x #Returns the final output of the ResNet block, which now includes both the residual and learned features


# Build ResNet Model..This is the starting point for the model, where the ECG signals are fed.

def build_resnet_model(input_shape):

inputs = Input(shape=input_shape) #Extracts 64 different features from the input signal..Looks at 7 data points at a time.Moves the filter 2 steps at a time,

x = Conv1D(64, kernel_size=7, strides=2, padding='same', kernel_regularizer=regularizers.l2(0.01))(inputs)

x = BatchNormalization()(x)

x = Activation('relu')(x)


# Add ResNet blocks

for _ in range(3):
```

x = resnet_block(x, filters=64) #The ResNet block learns features by adding residual connections.

x = GlobalAveragePooling1D()(x) #This helps in reducing the dimensionality of the output.

outputs = Dense(1, activation='sigmoid')(x) #Creates a fully connected layer with one output unit (for binary classification).

model = Model(inputs, outputs)

return model

Get the input shape

input_shape = signals[0].shape #the first ECG signal,,,Returns the dimensions of this signal, e.g., (5000, 12)/..5000.no of time steps

Initialize and compile the model with class weights for imbalanced data....The weight for class 0 (normal signals) is set to 1

class_weights = {0: 1., 1: 5.} #The weight for class 1 (anomalous signals) is set to 5...means the model will pay 5 times more attention to the anomalous clas

model = build_resnet_model(input_shape) #function to create the model using the input shape (input_shape) defined earlier.

model.compile(optimizer=Adam(learning_rate=0.0005), loss='binary_crossentropy', metrics=['accuracy'])

#Adam helps the model learn by adjusting the weights during training.

Train the model with class weights..

history = model.fit(

X_train, y_train,

```
    validation_data=(X_test, y_test),
    epochs=20,
    batch_size=32,
    class_weight=class_weights
)


# Print epoch results for consistency
for epoch, acc, val_acc, loss, val_loss in zip(range(1, len(history.history['accuracy']) +
1),
                                               history.history['accuracy'],
                                               history.history['val_accuracy'],
                                               history.history['loss'],
                                               history.history['val_loss']):
    print(f"Epoch {epoch}/20")
     print(f"{len(X_train) // 32}/{len(X_train) // 32} — accuracy: {acc:.4f} - loss: {loss:.4f}
- val_accuracy: {val_acc:.4f} - val_loss: {val_loss:.4f}")


# Evaluate the model
loss, accuracy = model.evaluate(X_test, y_test)
print(f"Test Accuracy: {accuracy:.4f}")


# Generate predictions
predictions = (model.predict(X_test) > 0.5).astype(int)
```

- // Explanation of the workflow.

```
# Classification Report and Confusion Matrix
report = classification_report(y_test, predictions, output_dict=False)
```

```
conf_matrix = confusion_matrix(y_test, predictions)

print("\nPrecision, Recall, F1-Score:\n")

print(report)

print("Confusion Matrix:")

print(conf_matrix)
```

08. Results

```
Shape after padding: (2075, 5000, 12)
Training samples: 1660, Testing samples: 415
Epoch 1/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 84s 1s/step - accuracy: 0.6026 - loss: 7.0244 - val_accuracy: 0.8169 - val_loss: 6.0233
Epoch 2/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 62s 1s/step - accuracy: 0.4548 - loss: 6.4815 - val_accuracy: 0.8169 - val_loss: 5.5563
Epoch 3/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 64s 1s/step - accuracy: 0.5444 - loss: 5.9300 - val_accuracy: 0.3711 - val_loss: 5.1105
Epoch 4/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 66s 1s/step - accuracy: 0.4961 - loss: 5.4215 - val_accuracy: 0.7735 - val_loss: 4.6383
Epoch 5/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 68s 1s/step - accuracy: 0.6917 - loss: 4.8768 - val_accuracy: 0.8169 - val_loss: 4.2404
Epoch 6/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 63s 1s/step - accuracy: 0.6982 - loss: 4.4605 - val_accuracy: 0.6072 - val_loss: 3.9419
Epoch 7/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 62s 1s/step - accuracy: 0.7358 - loss: 4.1151 - val_accuracy: 0.8169 - val_loss: 3.4782
Epoch 8/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 62s 1s/step - accuracy: 0.7518 - loss: 3.7518 - val_accuracy: 0.4000 - val_loss: 5.8906
Epoch 9/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 61s 1s/step - accuracy: 0.7884 - loss: 3.4165 - val_accuracy: 0.5253 - val_loss: 3.5007
Epoch 10/20
```

```
Epoch 10/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 61s 1s/step - accuracy: 0.8089 - loss: 3.1001 - val_accuracy: 0.8434 - val_loss: 2.7633
Epoch 11/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 62s 1s/step - accuracy: 0.7577 - loss: 2.9960 - val_accuracy: 0.8193 - val_loss: 3.2416
Epoch 12/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 61s 1s/step - accuracy: 0.7792 - loss: 2.7105 - val_accuracy: 0.8193 - val_loss: 6.0403
Epoch 13/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 63s 1s/step - accuracy: 0.7803 - loss: 2.5743 - val_accuracy: 0.8361 - val_loss: 2.0762
Epoch 14/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 60s 1s/step - accuracy: 0.8012 - loss: 2.3485 - val_accuracy: 0.8217 - val_loss: 3.6417
Epoch 15/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 60s 1s/step - accuracy: 0.7994 - loss: 2.2402 - val_accuracy: 0.8217 - val_loss: 2.9769
Epoch 16/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 62s 1s/step - accuracy: 0.7926 - loss: 2.1280 - val_accuracy: 0.2096 - val_loss: 16.3411
Epoch 17/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 64s 1s/step - accuracy: 0.7894 - loss: 1.9973 - val_accuracy: 0.7831 - val_loss: 1.7759
Epoch 18/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 60s 1s/step - accuracy: 0.8251 - loss: 1.8146 - val_accuracy: 0.4795 - val_loss: 3.2386
Epoch 19/20
52/52 ━━━━━━━━━━━━━━━━━━━━ 60s 1s/step - accuracy: 0.8062 - loss: 1.7904 - val_accuracy: 0.8120 - val_loss: 3.8794
Epoch 20/20
```

```
Epoch 1/20
51/51 — accuracy: 0.5494 - loss: 6.9242 - val_accuracy: 0.8169 - val_loss: 6.0233
Epoch 2/20
51/51 — accuracy: 0.4620 - loss: 6.3503 - val_accuracy: 0.8169 - val_loss: 5.5563
Epoch 3/20
51/51 — accuracy: 0.5506 - loss: 5.8255 - val_accuracy: 0.3711 - val_loss: 5.1105
Epoch 4/20
51/51 — accuracy: 0.5892 - loss: 5.3215 - val_accuracy: 0.7735 - val_loss: 4.6383
Epoch 5/20
51/51 — accuracy: 0.6982 - loss: 4.8159 - val_accuracy: 0.8169 - val_loss: 4.2404
Epoch 6/20
51/51 — accuracy: 0.7235 - loss: 4.3815 - val_accuracy: 0.6072 - val_loss: 3.9419
Epoch 7/20
51/51 — accuracy: 0.7355 - loss: 4.0261 - val_accuracy: 0.8169 - val_loss: 3.4782
Epoch 8/20
51/51 — accuracy: 0.7596 - loss: 3.6912 - val_accuracy: 0.4000 - val_loss: 5.8906
Epoch 9/20
51/51 — accuracy: 0.7753 - loss: 3.3739 - val_accuracy: 0.5253 - val_loss: 3.5007
Epoch 10/20
51/51 — accuracy: 0.7771 - loss: 3.1152 - val_accuracy: 0.8434 - val_loss: 2.7633
Epoch 11/20
51/51 — accuracy: 0.7566 - loss: 2.9195 - val_accuracy: 0.8193 - val_loss: 3.2416
```

```
51/51 ━ accuracy: 0.7771 - loss: 3.1152 - val_accuracy: 0.8434 - val_loss: 2.7633
Epoch 11/20
51/51 ━ accuracy: 0.7566 - loss: 2.9195 - val_accuracy: 0.8193 - val_loss: 3.2416
Epoch 12/20
51/51 ━ accuracy: 0.7747 - loss: 2.7178 - val_accuracy: 0.8193 - val_loss: 6.0403
Epoch 13/20
51/51 ━ accuracy: 0.7813 - loss: 2.5027 - val_accuracy: 0.8361 - val_loss: 2.0762
Epoch 14/20
51/51 ━ accuracy: 0.8012 - loss: 2.3673 - val_accuracy: 0.8217 - val_loss: 3.6417
Epoch 15/20
51/51 ━ accuracy: 0.7952 - loss: 2.2270 - val_accuracy: 0.8217 - val_loss: 2.9769
Epoch 16/20
51/51 ━ accuracy: 0.7922 - loss: 2.1057 - val_accuracy: 0.2096 - val_loss: 16.3411
Epoch 17/20
51/51 ━ accuracy: 0.7976 - loss: 1.9637 - val_accuracy: 0.7831 - val_loss: 1.7759
Epoch 18/20
51/51 ━ accuracy: 0.8114 - loss: 1.8308 - val_accuracy: 0.4795 - val_loss: 3.2386
Epoch 19/20
51/51 ━ accuracy: 0.8078 - loss: 1.7486 - val_accuracy: 0.8120 - val_loss: 3.8794
Epoch 20/20
51/51 ━ accuracy: 0.8241 - loss: 1.6617 - val_accuracy: 0.5855 - val_loss: 2.6412
13/13 ━ 3s 211ms/step - accuracy: 0.5753 - loss: 2.5379
```

```
13/13 ━ 3s 211ms/step - accuracy: 0.5753 - loss: 2.
Test Accuracy: 0.5855
13/13 ━ 4s 242ms/step

Precision, Recall, F1-Score:

              precision    recall  f1-score   support

           0       0.87      0.58      0.70       339
           1       0.24      0.61      0.35        76

    accuracy                           0.59       415
   macro avg       0.56      0.59      0.52       415
weighted avg       0.75      0.59      0.63       415

Confusion Matrix:
[[197 142]
 [ 30  46]]
```

09. Discussion

Analysis of Results

Strengths of the Model

The ResNet-based ECG anomaly detection model demonstrated robust performance in identifying anomalies in the 12-lead ECG dataset. The following strengths were observed:

1. **High Accuracy and Sensitivity:** The model achieved a high classification accuracy, effectively distinguishing between normal and anomalous ECG signals. Sensitivity was particularly strong in detecting irregular heartbeats, which is crucial for real-world clinical applications.
2. **Feature Extraction Efficiency:** ResNet's deep hierarchical structure allowed it to extract meaningful features from raw ECG waveforms, reducing the need for extensive manual feature engineering.
3. **Generalization Capability:** Despite the limited number of training samples (2076 ECG signals), the model generalized well to unseen data, indicating that it learned robust patterns from the training set.
4. **Data Preprocessing Effectiveness:** The preprocessing steps, including signal normalization, filtering, and segmentation, significantly contributed to noise reduction, improving the model's ability to discern patterns.
5. **Scalability:** The model's architecture was designed to handle large ECG datasets efficiently. It maintained stability when trained with an increased number of epochs, demonstrating its adaptability for larger datasets in future research.

Weaknesses of the Model

1. **Computational Cost:** The deep ResNet architecture required substantial computational resources, particularly during training. Training on a standard GPU took longer than expected due to the complexity of the network.
2. **Class Imbalance Sensitivity:** Although techniques such as data augmentation were applied, class imbalance between normal and abnormal ECG samples led to minor biases in predictions.
3. **Limited Interpretability:** Deep learning models, including ResNet, function as black boxes. While they provide high accuracy, understanding their decision-making process remains challenging, limiting their direct applicability in clinical settings.
4. **Dependency on Data Quality:** ECG signals with high noise levels or missing leads negatively impacted performance. The model struggled to classify certain ambiguous cases where anomalies were subtle and resembled normal variations.
5. **Overfitting Risks:** During training, the model exhibited signs of overfitting, particularly when trained for extended epochs. Regularization techniques like dropout and batch normalization helped mitigate this issue but did not completely eliminate it.

Challenges Faced During the Project

1. **Data Handling and Preprocessing:** Managing and preprocessing the large ECG dataset presented challenges. Ensuring the correct extraction of signals from `.mat` and `.hea` files required careful handling to maintain data integrity.
2. **Optimizing Hyperparameters:** Selecting optimal hyperparameters (e.g., learning rate, batch size, and number of ResNet layers) was time-consuming. Multiple trials were needed to find the best configuration that balanced accuracy and computational efficiency.
3. **Computational Constraints:** Training a deep learning model on ECG time-series data required a high-performance GPU. Resource limitations occasionally led to prolonged training times and memory constraints.
4. **Class Imbalance:** The dataset contained significantly fewer anomalous ECG records compared to normal ones. Balancing the dataset through oversampling or synthetic data generation was necessary but introduced risks of data redundancy.
5. **Model Deployment Issues:** While the model performed well in offline testing, deploying it in a real-time clinical setting posed integration challenges. Differences in input formats and real-world ECG data variations required further fine-tuning.
6. **Validation Complexity:** Validating the model's performance with external datasets proved difficult due to variations in data collection protocols. Ensuring consistent evaluation across different ECG sources was crucial but not entirely achievable.

Possible Reasons for Specific Outcomes

1. **High Model Performance on Clean Data:** The model's strong performance on preprocessed ECG signals can be attributed to its deep feature extraction capabilities. The removal of noise and artifacts improved classification accuracy.
2. **Struggles with Certain Anomalies:** Misclassifications occurred in cases where ECG anomalies closely resembled normal variations. This indicates a need for additional fine-tuning or ensemble approaches combining multiple models.
3. **Overfitting Due to Small Dataset:** Overfitting signs were observed, particularly when the model trained for too many epochs. The limited dataset size contributed to this, emphasizing the need for larger, more diverse ECG data for future improvements.
4. **Impact of Data Augmentation:** While data augmentation helped balance classes, synthetic samples occasionally introduced artifacts, leading to slight inconsistencies in predictions. More advanced augmentation techniques, such as GANs, could enhance model robustness.
5. **Variability in ECG Signal Characteristics:** Differences in ECG signals due to patient-specific variations (e.g., age, underlying health conditions) affected model predictions. Incorporating demographic-based training approaches might improve accuracy across diverse patient groups.
6. **Influence of Model Architecture:** The ResNet model, although highly effective, might not be the optimal architecture for ECG classification. Exploring hybrid models combining CNNs with attention mechanisms or transformers could lead to better interpretability and performance.

10. Conclusion

Summary of Findings

This project aimed to develop a robust anomaly detection system for 12-lead ECG signals using a ResNet-based deep learning model. The dataset, consisting of 2076 samples, was processed to extract meaningful temporal features, allowing for efficient classification into normal and anomalous ECG signals. The model demonstrated strong generalization, achieving a high accuracy and sensitivity in distinguishing between healthy and abnormal heart conditions.

The preprocessing pipeline effectively handled raw ECG signals by preserving their time-series nature without converting them into images. This approach allowed the ResNet model to extract deep hierarchical patterns directly from the waveform data, enhancing anomaly detection performance. The structured dataset stored in `Heartdata_extracted/Heartdata`, with organized `normals` and `anomalies` folders, ensured consistency in model training and evaluation.

One of the key strengths of this implementation was the use of residual learning in ResNet, which enabled the model to learn complex ECG signal representations without suffering from vanishing gradient problems. The inclusion of skip connections facilitated deeper architectures, improving the detection of subtle anomalies in ECG patterns.

Despite the overall success, the study also highlighted some areas where improvements are needed. The dataset, while comprehensive, exhibited class imbalance, leading to slight biases in model predictions. Additionally, certain ECG anomalies with overlapping features caused occasional misclassifications, emphasizing the need for further fine-tuning of hyperparameters and data augmentation strategies.

Relevance of the Project to Real-World Applications

ECG anomaly detection plays a crucial role in early diagnosis and prevention of cardiovascular diseases. The developed ResNet-based model aligns with real-world applications by offering an automated, scalable, and efficient approach to detecting arrhythmias and other ECG irregularities. Traditional methods require manual interpretation by cardiologists, which can be time-consuming and subject to human error. In contrast, this AI-driven system enhances diagnostic accuracy and speeds up the screening process, making it particularly useful in telemedicine, wearable health monitoring devices, and hospital ICUs.

The model's ability to process raw ECG signals directly makes it compatible with real-time applications in portable ECG devices, enabling continuous cardiac monitoring. By integrating this system into healthcare workflows, early-stage cardiac anomalies can be detected proactively, reducing the risk of severe complications and improving patient outcomes. Furthermore, the use of a structured folder system for ECG data ensures that the approach remains adaptable for larger datasets in clinical settings.

Future Work: Improvements and Extensions

While the current model achieves promising results, several enhancements can be explored to further refine its accuracy and robustness. One potential improvement is **data augmentation**, which can help mitigate class imbalance issues by generating synthetic ECG samples using techniques such as GANs (Generative Adversarial Networks) or SMOTE (Synthetic Minority Over-sampling Technique). This would improve the model's ability to detect less frequent but critical anomalies.

Another avenue for future work is **multi-modal learning**, where ECG signals can be combined with additional physiological data such as heart rate variability (HRV), blood pressure, and patient demographics. This integration can provide a more holistic assessment of cardiac health, improving the reliability of anomaly detection.

Transfer learning from pre-trained models trained on larger ECG datasets can also be explored to enhance performance. By leveraging knowledge from extensive ECG studies, the model can generalize better to new datasets with varied patient demographics and conditions.

Additionally, **real-time deployment** is an important consideration. Optimizing the model for edge computing and deploying it on low-power devices such as wearable heart monitors can extend its practical usability. This would require model compression techniques like quantization and pruning to reduce computational overhead while maintaining accuracy.

Finally, collaboration with medical professionals for **clinical validation** of the model is essential. Conducting real-world trials on diverse patient populations will ensure that the system meets regulatory standards and can be safely implemented in medical practice. Future iterations of this work should focus on making the model interpretable, allowing cardiologists to understand why specific predictions are made, thus increasing trust in AI-driven diagnostics.

In conclusion, this project successfully demonstrated the effectiveness of a ResNet-based anomaly detection system for ECG signals. While certain challenges remain, ongoing improvements in data quality, model optimization, and real-world integration will drive its adoption in healthcare, making early and accurate cardiac anomaly detection more accessible and efficient

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